

Algorithmic Pseudocode Sample 38

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

```
1:  $D_{train} \leftarrow$  Initialize training set
2: while  $C(fake) > threshold$  do ▷ The critic can be optimized until  $C(fake)$  is near 0.
3:   Randomly select a discrete variable  $c$  with equal probability
4:   Compute the probability mass function (PMF) of  $c$ 
5:   Randomly select a state  $i^*$  inherent to  $c$  according its PMF
6:   Create the conditional vector  $cond$  so that  $\sum_i cond(i) = 1$  and  $cond(i^*) = 1$ 
7:   for  $batch \in \{1, \dots, N_{batches}\}$  do ▷ Gradient descent with mini-batch
8:      $real \leftarrow d(c_{i^*} = 1) \sim D_{train}$  ▷ Sample batch of real examples respecting the constraint
9:      $z \sim \mathcal{N}(0, 1)$  ▷ Sample noise
10:     $fake \leftarrow \tilde{d} \sim G(z)$  ▷ Sample fake examples
11:     $real \leftarrow [real] \times 10$  ▷ Stack input 10 times for Pac configuration
12:     $fake \leftarrow [fake] \times 10$ 
13:     $L^j \leftarrow (C(fake_j) - C(real_j)) + CE(\tilde{c}, cond)$ 
14:     $L^{batch} \leftarrow L^{batch} + \lambda(\|\nabla L^{batch}\|_2 - 1)^2$  ▷ Apply gradient penalty
15:     $w_{crit} \leftarrow w_{crit} + Adam(\nabla_{w_{crit}} \frac{1}{m} \sum_i L^{batch}(i))$  ▷ Updating  $C$  with Adam
16:    if  $batch \bmod k = 0$  then ▷ Synchronicity, depends on  $k$ 
17:       $w_{gen} \leftarrow w_{gen} + Adam(\nabla_{w_{gen}} \frac{1}{m} \sum_i -C(G(z)))$  ▷ Updating  $G$  with Adam
18:    end if
19:  end for
20: end while
```
